

Binary Relations

1 Binary relations

Let S be a set. A binary relation (or simply a relation) $\mathcal{R}$ on S is any subset Γ of $S \times S$. One usually writes $x\mathcal{R}y$ when $(x, y) \in \Gamma$, and we say that x is in relation with y . Γ is called the graph of $\mathcal{R}$.

Example 1

1. The subset Γ_1 defined by $\Gamma_1 = \{(1, 2), (1, 3), (2, 1)\}$ is a binary relation on $S = \{1, 2, 3\}$.
2. The subset Γ_2 of $\mathbb{R}^2$, defined by $\Gamma_2 = \{(x, y) \in \mathbb{R}^2 : x < y\}$, is a binary relation on $\mathbb{R}$.

Intuitively, a relation $\mathcal{R}$ on a set S is a property that concerns couples (x, y) of elements from S . Let $\mathcal{R}$ be a binary relation on S .

- $\mathcal{R}$ is said to be reflexive if for any $x \in S$, $x\mathcal{R}x$.
- $\mathcal{R}$ is said to be irreflexive if for any $x \in S$, non $x\mathcal{R}x$.
- $\mathcal{R}$ is said to be symmetric if for any $x, y \in S$, $x\mathcal{R}y$ implies $y\mathcal{R}x$.
- $\mathcal{R}$ is said to be asymmetric if for any $x, y \in S$, $x\mathcal{R}y$ implies non $y\mathcal{R}x$.
- $\mathcal{R}$ is said to be antisymmetric if for any $x, y, z \in S$, $x\mathcal{R}y$ and $y\mathcal{R}x$ imply $x = y$.
- $\mathcal{R}$ is said to be transitive if for any $x, y, z \in S$, $x\mathcal{R}y$ and $y\mathcal{R}z$ imply $x\mathcal{R}z$.
- $\mathcal{R}$ is said to be circular if for any $x, y, z \in S$, $x\mathcal{R}y$ and $y\mathcal{R}z$ imply $z\mathcal{R}x$.

Example 2

1. The relation represented above by Γ_1 is not reflexive, not symmetric, not antisymmetric, not transitive, not circular.
2. The relation represented above by Γ_2 is not reflexive, not symmetric, antisymmetric, transitive.

3. Equality between elements of a set is a binary relation that is reflexive, symmetric, antisymmetric, and transitive.
4. The trivial relation on a set S is the relation where every element is related to every other element. Its graph is $S \times S$ and it is reflexive, symmetric, transitive, and circular.
5. The empty relation, is the relation where a given element is related to none. It is not reflexive, it is symmetric, it is transitive.
6. Perpendicularity is a binary relation on the set of lines in a plan. It is symmetric, not reflexive, not transitive, and not circular.
7. Parallelism is a binary relation on the set of lines in a plan. It is reflexive, symmetric, transitive, and circular.
8. Let S be a set. On $\mathcal{P}(S)$, the relation $\mathcal{R}$ defined by: $A\mathcal{R}B$ if and only if $A = B$ or $A = \overline{B}$, is reflexive, symmetric, and transitive.
9. On $\mathbb{R}$, the binary relation $\mathcal{R}$ defined by: $x\mathcal{R}y$ if and only if $x \leq 2^y$ is neither symmetric nor antisymmetric.
10. On $\mathbb{R}$, the relation $\mathcal{R}$ defined by: $x\mathcal{R}y$ if and only if $x^2 = y^2$ is symmetric but not antisymmetric.

► Note that:

- The notion of symmetric and antisymmetric are not opposite (are not contrary to each other).
- Equality is both symmetric and antisymmetric at the same time.
- Consider a relation $\mathcal{R}$ on a set E . If $\left\{ \begin{array}{l} \mathcal{R} \text{ is antisymmetric} \\ \text{and} \\ \exists x_0, y_0 \in E, x_0 \neq y_0, x_0\mathcal{R}y_0, \end{array} \right.$ then $\mathcal{R}$ is not symmetric.

1.1 Equivalence relation

Let S be a set. A binary relation $\mathcal{R}$ on S is said to be an equivalence relation if it satisfies the following conditions:

- $\mathcal{R}$ is reflexive,
- $\mathcal{R}$ is symmetric,
- $\mathcal{R}$ is transitive.

Example 3

1. Equality is an equivalence relation on any set S .

2. Equipotence is an equivalence relation between sets.
3. Parallelism is an equivalence relation on the set of lines in a plan.
4. On $\mathbb{N} \times \mathbb{Z}^*$, the relation defined by:

$$(a, b) \mathcal{R} (c, d) \text{ if and only if } a + b = c + d$$

is an equivalence relation.

5. On $\mathbb{R}$, the relation $\mathcal{R}$ defined by:

$$x \mathcal{R} y \text{ if and only if } x^2 - y^2 = x - y$$

is an equivalence relation.

1.1.1 Equivalence classes and quotient set

Let $\mathcal{R}$ be an equivalence relation on a set S and let $x \in S$. The set of all elements y belonging to S such that $x \mathcal{R} y$ is called the equivalence class of x and is denoted by $\bar{x}$ or $cl_{\mathcal{R}}(x)$. That is,

$$\bar{x} = \{y \in S : x \mathcal{R} y\}.$$

The set of all equivalence classes $\{\bar{x} : x \in S\}$ is called the quotient set associated with the relation $\mathcal{R}$ and is denoted by $S/\mathcal{R}$. That is,

$$S/\mathcal{R} = \{\bar{x} : x \in S\}.$$

Proposition 4 *Let $\mathcal{R}$ be an equivalence relation on a set S . The set of classes of $\mathcal{R}$ is a partition of S .*

Proof. Let $x \in S$. By the reflexivity of $\mathcal{R}$, $x \mathcal{R} x$, which implies that x belongs to the equivalence class of x . Therefore, every equivalence class is nonempty. To show that S is the union of all equivalence classes, observe that every element $y \in S$ belongs to some equivalence class, as each element y is related to itself. To prove that the union of the equivalence classes is disjoint, consider two equivalence classes $\bar{x}$ and $\bar{y}$. We need to show that they are either disjoint or identical. Suppose $\bar{x} \cap \bar{y} \neq \emptyset$. Let $z \in \bar{x} \cap \bar{y}$. Since $z \in \bar{x}$, we have $x \mathcal{R} z$. Similarly, since $z \in \bar{y}$, we have $y \mathcal{R} z$. By the transitivity of $\mathcal{R}$, we have $x \mathcal{R} z$ and $z \mathcal{R} y$, then $x \mathcal{R} y$. Now, consider any $x' \in \bar{x}$. Since $x' \mathcal{R} x$ and $x \mathcal{R} y$, by transitivity $x' \mathcal{R} y$. Thus, $x' \in \bar{y}$, which implies that $\bar{x} \subseteq \bar{y}$. Similarly, any $y' \in \bar{y}$ satisfies $y' \mathcal{R} y$ and $y \mathcal{R} x$, so $y' \mathcal{R} x$ by transitivity. Thus, $y' \in \bar{x}$, which implies $\bar{y} \subseteq \bar{x}$. Therefore, $\bar{x} = \bar{y}$. This shows that equivalence classes are disjoint or identical, establishing that S is the disjoint union of the equivalence classes of $\mathcal{R}$. ■

Proposition 5 *Let P be a partition of a set S . Then there exists a unique equivalence relation $\mathcal{R}_P$ such that the quotient set is P .*

Proof. *Let P be a partition of S . We define $\mathcal{R}_P$ in S as follows:*

$$x\mathcal{R}_Py \text{ if and only if } \exists A \in P \text{ such that } x, y \in A.$$

(In other words, x, y are in relation if and only if they are lying in the same subset belonging to P).

The relation $\mathcal{R}_P$ is reflexive: Because P is a partition, then for any $x \in S$, there must exists $A \in P$ such that $x \in A$, therefore $x\mathcal{R}_Px$.

The relation $\mathcal{R}_P$ is symmetric: Let $x, y \in S$ such that $x\mathcal{R}_Py$. Then, $x, y \in A$ for some $A \in P$ so $y, x \in A$, hence $y\mathcal{R}_Px$.

The relation $\mathcal{R}_P$ is transitive: Let $x, y, z \in S$ such that $x\mathcal{R}_Py$ and $y\mathcal{R}_Pz$. Then $\exists A_1 \in P$ with $x, y \in A_1$ and $\exists A_2 \in P$ with $y, z \in A_2$. Here we have $y \in A_1 \cap A_2$. But A_1, A_2 are elements of the partition P . Therefore $A_1 = A_2$ and since $x \in A_1$ and $A_1 = A_2$ we have $x \in A_2$, so $x, z \in A_2$ and $x\mathcal{R}_Pz$. This shows that $\mathcal{R}_P$ is an equivalence relation and $\bar{x} = \{y \in S \mid x\mathcal{R}_Py\} = A$. So $S/\mathcal{R}_P = P$. It is obvious that $\mathcal{R}_P$ is the unique equivalence relation S such that $P = S/\mathcal{R}_P$. ■

We conclude from the previous two propositions that there is a canonical bijection between equivalence relations on set S and partitions of S .

Example 6

1. *Let S be a set. The equality is an equivalence relation. $S/\mathcal{R}$ is S itself.*
2. *Let n be a positive integer, and consider the equivalence relation $\mathcal{R}$ defined on $\mathbb{Z}$ by:*

$$x\mathcal{R}y \text{ if and only if } \exists k \in \mathbb{Z} \text{ such that } x - y = nk.$$

Then, the quotient set is $\mathbb{Z}/\mathcal{R} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$, which we denote usually by $\mathbb{Z}/n\mathbb{Z}$ or $\mathbb{Z}_n$.

3. *Let $S = \mathbb{R}^2$. The relation $\mathcal{R}$ on $\mathbb{R}^2$ defined by:*

$$(x_1, y_1)\mathcal{R}(x_2, y_2) \text{ if and only if } \|(x_1, y_1)\| = \|(x_2, y_2)\|,$$

where $\|(x, y)\| = \sqrt{x^2 + y^2}$, is an equivalence relation. Here,

$\mathbb{R}^2/\mathcal{R}$ is the the set of circles centred at $(0, 0)$.

4. *Let $S = \mathbb{R}^2$. The relation $\mathcal{R}$ on $\mathbb{R}^2$ defined by:*

$$(x_1, y_1)\mathcal{R}(x_2, y_2) \text{ if and only if } x_1 = x_2$$

is an equivalence relation. Here,

$\mathbb{R}^2/\mathcal{R}$ is the the set of vertical lines.

5. Let $S = \mathbb{R}^2$. The relation $\mathcal{R}$ on $\mathbb{R}^2$ defined by:

$$(x_1, y_1) \mathcal{R} (x_2, y_2) \text{ if and only if } y_1 = y_2$$

is an equivalence relation. Here,

$$\mathbb{R}^2 / \mathcal{R} \text{ is the set of horizontal lines.}$$

1.2 Orders on a set

- Let S be a set and $\mathcal{R}$ a binary relation on S . The relation $\mathcal{R}$ is said to be a partial order (or poset) if it satisfies the following conditions:

1. $\mathcal{R}$ is reflexive,
2. $\mathcal{R}$ is antisymmetric,
3. $\mathcal{R}$ is transitive.

A set equipped with a partial order relation is called an ordered set. We often use $\leq$ to denote a partial order relation.

- A binary relation $\mathcal{R}$ on a set S is said to be a total order if:

1. $\mathcal{R}$ is a partial order,
2. For any $x, y \in S$, either $x \mathcal{R} y$ or $y \mathcal{R} x$ (i.e., any pair of elements of S is comparable under $\mathcal{R}$).

- A binary relation $\mathcal{R}$ over set S is said to be a strict order if $\mathcal{R}$ is irreflexive, asymmetric and transitive. Most often, a strict order on S is denoted by the symbol $<$.

- A binary relation $\mathcal{R}$ on a set S is said to be a preorder if it is reflexive and transitive.

Example 7

1. Let $S = \{a, b, c\}$ and let $\Gamma = \{(a, a), (b, b), (c, c), (a, b), (c, b)\}$. The relation whose graph is Γ is a partial order relation.
2. Equality is a partial order on any set S .
3. The division on $\mathbb{N}$, denoted $/$, is a partial order.
4. The relation less or equal to, denoted $\leq$, is a total order on $\mathbb{R}$.
5. The relation less than to, denoted $<$, is a strict order on $\mathbb{R}$.
6. The inclusion relation, denoted $\subset$ on the power set $\mathcal{P}(S)$ of a set S is a partial order.

7. On $\mathbb{N}$, the relation $\mathcal{R}$, defined by:

$$a\mathcal{R}b \text{ if only if } b = 2^k a \text{ for some integer } k \geq 0$$

is a partial order.

There is only one relation that is simultaneously an equivalence relation and a partial order, and that is the equality relation. As shown earlier, equality is both an equivalence relation and a partial order. To prove that the only relation that is both an equivalence relation and a partial order is the equality relation, we need to show:

$$\forall x, y \in S, x\mathcal{R}y \iff x = y.$$

Let $x, y \in S$ and assume R is both an equivalence relation and a partial order.

Let $x, y \in S$. $x\mathcal{R}y$ gives $y\mathcal{R}x$ because $\mathcal{R}$ is symmetric then we have $x = y$ because $\mathcal{R}$ is antisymmetric. Conversely, suppose $x = y$. Since $\mathcal{R}$ is reflexive we have $x\mathcal{R}y$ since $y = x$.

1.2.1 Order on Cartesian product

Let S be an ordered set. There are usually two ways to define an order on S^2 :

- The coordinate order, defined by:

$$(\alpha, \beta) \leq (u, v) \iff \alpha \leq u \text{ and } \beta \leq v.$$

This is the Cartesian product order.

- The lexicographic order, defined by:

$$(\alpha, \beta) \leq (u, v) \iff \alpha \leq u \text{ or } (\alpha = u \text{ and } \beta \leq v).$$

Note that the lexicographic order, which also can be defined on S^n ($n \geq 3$), compares the first differing coordinates, whereas the coordinate order on S^n compares all coordinates.